

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA0503

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : Case Study (Medium)

Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	5
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	5
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	5
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	5
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	5
6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	5
7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	5

8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	5
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	5
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	5

Code of Conduct:

I, JASWANTH (192525345) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: I.JASWANTH

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions	Logical solutions with	Basic solutions with weak	Incorrect or no solution provided
		with strong justification	adequate justification	justification	

Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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ANSWERS:

1. Online Shopping – 3NF

START

Identify FDs:

OrderID → CustID

CustID → CustName

ProdID → ProdName, Price

(OrderID, ProdID) → Qty

Create Customer(CustID, CustName)

Create Product(ProdID, ProdName, Price)

Create Orders(OrderID, CustID)

Create OrderDetails(OrderID, ProdID, Qty)

```
mysql> SELECT O.OrderID, C.CustName, P.ProdName, OD.Qty, P.Price
-> FROM Orders O
-> JOIN Customer C ON O.CustID=C.CustID
-> JOIN OrderDetails OD ON O.OrderID=OD.OrderID
-> JOIN Product P ON OD.ProdID=P.ProdID;
+-----+-----+-----+-----+-----+
| OrderID | CustName | ProdName | Qty | Price |
+-----+-----+-----+-----+-----+
|      1 | Ravi    | Laptop  | 1   | 50000 |
|      1 | Ravi    | Mouse   | 2   | 800   |
+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

END

2. Hospital – BCNF

START

Identify FDs:

PID → PName, DocID, WardNo

DocID → DocName, Specialization

WardNo → WardName

Create Patient(PID, PName, DocID, WardNo)

Create Doctor(DocID, DocName, Specialization)

Create Ward(WardNo, WardName)

```
mysql> SELECT P.PID, P.PName, D.DocName, D.Specialization, W.WardName
-> FROM Patient P
-> JOIN Doctor D ON P.DocID=D.DocID
-> JOIN Ward W ON P.WardNo=W.WardNo;
```

```
+-----+-----+-----+-----+-----+
| PID | PName | DocName | Specialization | WardName |
+-----+-----+-----+-----+-----+
| 101 | Ravi  | Kumar   | Cardiology     | General  |
| 102 | Sita  | Priya   | Neurology      | ICU      |
+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

END

3. University Enrollment – 2NF

START

Identify FDs:

SID → SName

CourseID → CourseName, Faculty

(SID, CourseID) → Grade

Create Student(SID, SName)

Create Course(CourseID, CourseName, Faculty)

Create Enrollment(SID, CourseID, Grade)

END

```
mysql> SELECT E.SID, S.SName, C.CID, C.CName, C.Faculty, E.Grade
-> FROM Enrollment E
-> JOIN Student S ON E.SID=S.SID
-> JOIN Course C ON E.CID=C.CID;
```

```
+-----+-----+-----+-----+-----+-----+
| SID | SName | CID | CName          | Faculty | Grade |
+-----+-----+-----+-----+-----+-----+
| 101 | Arun  | 201 | DBMS           | Kumar   | A     |
| 101 | Arun  | 202 | Operating Systems | Ravi    | B     |
| 102 | Priya | 201 | DBMS           | Kumar   | A     |
| 102 | Priya | 202 | Operating Systems | Ravi    | A     |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

4. Airline Booking – BCNF

START

Identify FDs:

FlightNo → DeptCity, ArrCity

PassID → PassName

(FlightNo, Date, Seat) → PassID

Create Flight(FlightNo, DeptCity, ArrCity)
 Create Passenger(PassID, PassName)
 Create Booking(FlightNo, PassID, Seat, Date)

END

```
mysql> SELECT B.FlightNo, B.PassID, P.PassName, B.Seat,
-> F.DeptCity, F.ArrCity, B.BookingDate
-> FROM Booking B
-> JOIN Passenger P ON B.PassID=P.PassID
-> JOIN Flight F ON B.FlightNo=F.FlightNo;
+-----+-----+-----+-----+-----+-----+-----+
| FlightNo | PassID | PassName | Seat | DeptCity | ArrCity | BookingDate |
+-----+-----+-----+-----+-----+-----+-----+
| AI101    | 1      | Ravi     | 12A  | Chennai  | Delhi   | 2026-08-20   |
| AI102    | 2      | Kiran    | 10B  | Mumbai   | Delhi   | 2026-08-21   |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

5. HR System – 3NF

START

Identify FDs:

EmpID → EmpName, DeptID, JobID

DeptID → DeptName, Location

JobID → JobTitle, SalaryGrade

Create Employee(EmpID, EmpName, DeptID, JobID)

Create Department(DeptID, DeptName, Location)

Create Job(JobID, JobTitle, SalaryGrade)

END

```
mysql> SELECT E.EmpID, E.EmpName, D.DeptName, D.Location,
-> J.JobTitle, J.SalaryGrade
-> FROM Employee E
-> JOIN Department D ON E.DeptID=D.DeptID
-> JOIN Job J ON E.JobID=J.JobID;
+-----+-----+-----+-----+-----+-----+
| EmpID | EmpName | DeptName | Location | JobTitle           | SalaryGrade |
+-----+-----+-----+-----+-----+-----+
| 1001  | Ravi    | IT       | Chennai  | Software Engineer  | G5          |
| 1002  | Sita    | HR       | Bangalore | HR Manager         | G7          |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

6. Library – 4NF

START

Identify MVDs:

BookID →→ Author

BookID →→ Genre

Create Book(BookID, Title)
 Create BookAuthor(BookID, Author)
 Create BookGenre(BookID, Genre)
 Create Member(MemberID, MemberName)
 Create Borrow(MemberID, BookID, BorrowDate)

END

```
mysql> SELECT B.BookID, B.Title, A.Author, G.Genre
-> FROM Book B
-> JOIN BookAuthor A ON B.BookID=A.BookID
-> JOIN BookGenre G ON B.BookID=G.BookID;
```

BookID	Title	Author	Genre
101	Database Systems	Korth	Database
101	Database Systems	Korth	Education
101	Database Systems	Ramakrishnan	Database
101	Database Systems	Ramakrishnan	Education

4 rows in set (0.01 sec)

7. Telecom Billing

START

Identify candidate keys:

CustomerID

Phone

BillID

Identify FDs:

CustomerID → CustomerName, Phone

Phone → CustomerID, CustomerName

BillID → CustomerID, Amount

Create Customer(CustomerID, CustomerName, Phone)

Create Bill(BillID, CustomerID, Amount)

END

```
mysql> SELECT B.BillID, C.CustomerID, C.CustomerName,
-> C.Phone, B.Amount
-> FROM Bill B JOIN Customer C ON B.CustomerID=C.CustomerID;
```

BillID	CustomerID	CustomerName	Phone	Amount
1001	101	Ravi	9876543210	599
1002	102	Kiran	9876543211	299

```
2 rows in set (0.01 sec)
```

8. Retail Inventory – Normalization

START

Identify FDs:

PID → PName, SupplierID, Category, Price

SupplierID → SupplierName

Create Product(PID, PName, SupplierID, Category, Price)

Create Supplier(SupplierID, SupplierName)

Create Warehouse(WarehouseID, WarehouseName)

END

```
mysql> SELECT P.PID, P.PName, S.SupplierName,
-> P.Category, P.Price, P.Warehouse
-> FROM Product P JOIN Supplier S ON P.SupplierID=S.SupplierID;
```

PID	PName	SupplierName	Category	Price	Warehouse
501	Laptop	ABC	Electronics	50000	Chennai
502	Mouse	XYZ	Accessories	800	Chennai

```
2 rows in set (0.01 sec)
```

9. School ERP – BCNF

START

Identify FDs:

StudentID → StudentName, DeptID

DeptID → DeptName

SubjectID → SubjectName, TeacherID

TeacherID → TeacherName

Create Student(StudentID, StudentName, DeptID)

Create Department(DeptID, DeptName)

Create Subject(SubjectID, SubjectName, TeacherID)

Create Teacher(TeacherID, TeacherName)

Create StudentSubject(StudentID, SubjectID, Marks)

END

```
mysql> SELECT S.StudentID, S.StudentName, D.DeptName,
-> SU.SubjectName, T.TeacherName, SS.Marks
-> FROM StudentSubject SS
-> JOIN Student S ON SS.StudentID=S.StudentID
-> JOIN Department D ON S.DeptID=D.DeptID
-> JOIN Subject SU ON SS.SubjectID=SU.SubjectID
-> JOIN Teacher T ON SU.TeacherID=T.TeacherID;

+-----+-----+-----+-----+-----+-----+
| StudentID | StudentName | DeptName | SubjectName | TeacherName | Marks |
+-----+-----+-----+-----+-----+-----+
| 1 | Ravi | Computer Science | DBMS | Kumar | 85 |
| 1 | Ravi | Computer Science | Operating Systems | Priya | 90 |
| 2 | Kiran | Computer Science | DBMS | Kumar | 78 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

10. Movie Streaming – 5NF

START

Identify Join Dependency:

Movie, Actor, Platform

Create Movie(MovieID, MovieName)

Create Actor(ActorID, ActorName)

Create Platform(PlatformID, PlatformName)

Create MovieActor(MovieID, ActorID)

Create MoviePlatform(MovieID, PlatformID)

Create ActorPlatform(ActorID, PlatformID)

Verify lossless join

END

```
mysql> SELECT M.MovieName, A.ActorName, P.PlatformName
-> FROM Movie M
-> JOIN MovieActor MA ON M.MovieID=MA.MovieID
-> JOIN Actor A ON MA.ActorID=A.ActorID
-> JOIN MoviePlatform MP ON M.MovieID=MP.MovieID
-> JOIN Platform P ON MP.PlatformID=P.PlatformID;

+-----+-----+-----+
| MovieName | ActorName | PlatformName |
+-----+-----+-----+
| RRR | NTR | Netflix |
| RRR | Ram Charan | Netflix |
| RRR | NTR | Prime Video |
+-----+-----+-----+
3 rows in set (0.01 sec)
```